

Speech-Driven Drafting of First Information Reports — Tamil Nadu Police

Turning a spoken complaint into a verifiable, BNS-ready FIR draft.

Team

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Guide

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A Spoken Complaint, a Fixed Form, a New Penal Code

1

Bilingual & colloquial

Complaints are spoken in Tamil, with heavy Tamil–English code-switching.

2

A new penal code

BNS 2023 replaced the IPC on 1 July 2024 — every section is renumbered.

3

The form is fixed

The TN IF-1 form is standardised, so drafting is slot-filling, not free writing.

Key Question: *Can we draft a verifiable FIR, with the right BNS sections, straight from speech?*

Goal: assist the officer — draft and justify, never decide.

Existing Work & the Research Gap

Speech & extraction

Whisper

IndicWhisper

MuRIL

IndicBERT

OpenNyAI NER

Legal reasoning & generation

ILSI / LeSICiN

InLegalBERT

ILDC

LegalBench

RAG

QLoRA

GAP

Strong components exist in isolation. None targets spoken, bilingual complaints; statute-ID data is IPC-era; and the explanations are shallow.

OUR APPROACH

A five-stage, officer-in-the-loop pipeline with explainable BNS statute identification.

Speech, Legal NLP & the Gap

1. Speech & Extraction

- **Whisper / IndicWhisper:** strong Tamil ASR, weak on code-switch (Radford 2023; Javed 2023).
- **MuRIL / IndicBERT:** Indian-language encoders (Khanuja 2021; Kakwani 2020).
- **OpenNyAI NER:** 14 legal entity types over judgments (Kalamkar 2022).

Built on read speech and judgments, not spoken complaints.

2. Statute ID & Generation

- **ILSI + LeSICiN:** 66K facts → IPC sections, graph model (Paul 2022).
- **InLegalBERT / ILDC / LegalBench:** legal reasoning (Paul 2023; Malik 2021; Guha 2023).
- **RAG + constrained decoding + QLoRA:** safe, local generation (Lewis 2020; Dettmers 2023).

IPC-era labels; explanations are shallow, not element-wise.

3. Research Gap

- No system targets spoken, bilingual, colloquial complaints.
- Statute-ID data is IPC-era; must be bridged to BNS 2023.
- Explanations are shallow, not an element-wise offence check.

Our Positioning: *integrate the components for TN FIRs, bridge IPC→BNS, and justify every section element-wise.*

Objectives & Contributions

1

Tamil / code-switched ASR with normalisation

2

Extract IF-1 fields with confidence + provenance

3

Explainable BNS statute identification

4

Cognizability gate — FIR vs CSR

5

Deterministic FIR fill + grounded narrative

6

Officer verification workspace + audit

EXPECTED CONTRIBUTION

An explainable, officer-in-the-loop system that drafts a BNS-ready Tamil Nadu FIR from a spoken complaint.

Datasets & Data Plan

STATUTE & LEGAL DATA

66K

ILSI facts (IPC-labelled)

46,545

OpenNyAI legal entities

358

BNS 2023 sections

IPC-era → bridged to BNS via a verified map.

SPEECH DATA

1,684 h

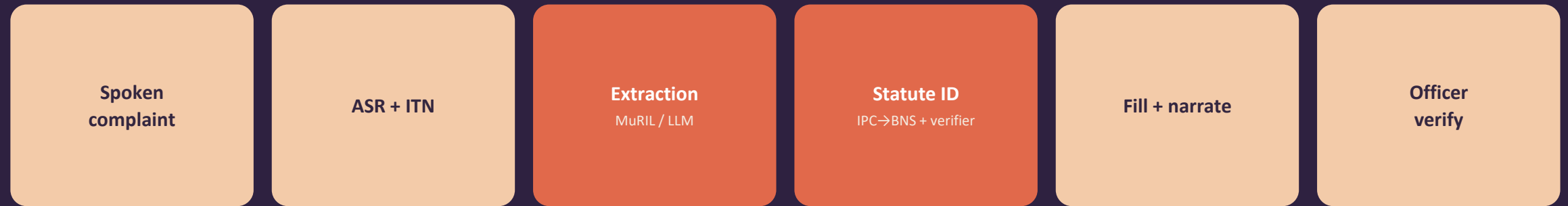
Kathbath labelled Tamil speech

3,335

FLEURS-ta evaluation clips

No public Tamil complaint corpus — so we go synthetic-first, with a self-recorded human gold set staged later.

Proposed Methodology



Listen → Extract → Justify → Draft → Verify

The form is fixed — the real work is understanding the complaint and choosing the right sections.

Models for a 16 GB GPU

BUDGET

16 GB

single consumer GPU

Resident 7B + int8 ASR + retriever co-reside in ~13–14 GB. PEFT keeps training in budget.

Stage	Model (precision)
ASR	Whisper large-v3 (int8) / IndicConformer
Extraction / verify / narrative	Qwen2.5-7B-Instruct (4-bit), resident
NER / classifier / retriever	MuRIL, InLegalBERT, BGE-m3
Data generation (offline)	Qwen2.5-14B (4-bit), GPU-alone

Local-only, offline-degradable — no dependency on hosted APIs.

Explainable Statute Identification

1

Classify offences from the narrative

2

Map IPC → BNS 2023 (verified)

3

Check each statutory ingredient

4

Cognizable? → FIR / CSR

302→103 · 304B→80 · 498A→85 · 3(5)

PURPOSE

Not just which sections — but why each one applies, element by element.

Existing models predict labels; we check each offence's ingredients against the complaint, with supporting spans, so an officer can verify and defend the charge.

Evaluation Plan

Speech

- WER, CER
- Code-switch WER
- Punctuation / ITN accuracy

Extraction & Statute

- Slot precision / recall / F1
- Micro / macro-F1 vs LeSiCiN
- Cognizability accuracy

End-to-end

- Field completeness
- Officer edit distance
- Time-to-draft
- Narrative faithfulness

Faithfulness ↔ Coverage benchmark statute ID against LeSiCiN on ILSI (BNS-mapped); every justification checked for entailment.

Architecture, Feasibility & Work Plan

Audio

ASR

Extract

Statute

IPC→BNS

Instantiate

Officer UI

Audit

Feasibility

- Open datasets
- Local 16 GB GPU
- PEFT + quantization
- DPDP-aligned, offline

Completed

Schema, verified datasets, and the IPC→BNS map (71/100 sections high-confidence) are done.

Next Steps

Statute-ID baseline → ASR on FLEURS-ta → synthetic data → eval harness.

Conclusion & Future Scope

OFFICER WORKSPACE (demo target)

Review UI

- Transcript ↔ fields with provenance
- Suggested sections + justifications
- Edit / override / immutable audit

FUTURE SCOPE — IF TIME PERMITS

Real complaint audio

More special acts

Guided-interview mode

On-device deployment

Drafting the form is the easy part. The contribution is understanding the complaint and justifying the law — with the officer in control.

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